

1 **Prototype Effects and Gradience in the Constructional Network:**
2 **Structural Priming Evidence from Caused-Motion and**
3 **Resultative Constructions in Mandarin *bǎ*-Sentence**

4
5 **Abstract:** The present study employed a structural priming paradigm with a maze self-paced reading
6 task to investigate the online processing of the caused-motion and resultative constructions within the
7 Chinese *bǎ*-construction, focusing on the directionality of cross-constructional priming. Results showed
8 that caused-motion primes significantly slowed resultative targets relative to unrelated primes, whereas
9 caused-motion targets showed no priming effect, revealing a unidirectional inhibitory pattern. Verb
10 repetition modulated this pattern construction-specifically. Verb overlap facilitated processing in caused-
11 motion-to-caused-motion pairs but tended to slow caused-motion-to-resultative pairs, with no reliable
12 effect at the verb position itself. These findings support a gradient, prototype-based organization of the
13 *bǎ*-construction network, in which the prototypical caused-motion construction exerts a stronger
14 directional influence on the semantically adjacent resultative construction, advancing constructional
15 network theory and providing psycholinguistic evidence for the psychological reality of constructional
16 relations in Mandarin Chinese.

17 **Keywords:** caused-motion; resultative; *bǎ*-construction; constructional network; structural priming
18

1. Introduction

Cognitive Construction Grammar claims that linguistic knowledge is organized as form-meaning pairings termed *constructions*, which are stored in the language user's mind via a structured network, known as the *construct-i-con* (Goldberg, 2003; Jurafsky, 1991). In the published literature, the structure of constructional network is typically categorized into two different dimensions: *vertical links* and *horizontal links*. The vertical links, also referred to *taxonomic links* or *inheritance links*, highlight hierarchical organization between abstract, higher-level constructions and their more specific, lower-level instantiations, with the latter inheriting both formal and semantic features from the former (Goldberg, 1995; Hilpert, 2014; Traugott & Trousdale, 2013). The horizontal links highlight connections established between constructions at the same abstract level based on formal and semantic similarity and usage frequency (Diessel, 2019, 2023; Norde & Morris, 2018; Sommerer & Smirnova, 2020; Ungerer & Hartmann, 2023). The two links differ not only in their organizational principles but also exhibit distinct psychological reality in language processing.

Recent empirical studies employing structural priming paradigms have demonstrated that different types of constructional relations possess distinct psychological realities. Research on both first and second language processing has provided increasing empirical evidence for the existence of both vertical inheritance and horizontal associative links within various constructional families (Bouso et al., 2024; Hong & Ma, 2024; Ungerer, 2021).

However, while these relations have been extensively verified in English, it remains unclear whether constructional relations in Mandarin Chinese, as an analytic language, displays similar differences between vertical inheritance and horizontal association at the level of cognitive processing. In this regard, the present study focuses on the caused-motion construction and the resultative construction within the *bǎi*-construction, drawing on the Maze Self-Paced Reading Task with a structural priming paradigm. The purpose of this study is to investigate the relations between these two constructions within the Chinese constructional network and to provide solid evidence

for Cognitive Construction Grammar concerning the directionality of cross-
constructional priming in Mandarin.

The remainder of the paper is organized as follows. Section 2 reviews the syntactic
and semantic properties of the *bǎ*-construction and relevant structural priming research.
Section 3 describes the experimental design, Section 4 reports the results, and Sections
5 and 6 discuss their theoretical implications for the organization of the constructional
network.

2. Research Background

2.1. Network Relations of the Chinese *bǎ*-construction

Chinese *bǎ*-construction employs *bǎ* (“把”) as a prepositional marker to front the
direct object before the verb, highlighting the agent’s control over the patient and
emphasizing the realization of the event’s resultant state (Huang et al., 2009; Li &
Thompson, 1989; Niu, 2008; Shi, 2010). In terms of semantic interpretation, current
research analyzes this construction from the perspectives of disposal or causation,
among which the causative-resultative framework has emerged as a highly explanatory
approach (Guo, 2003; Ye, 2004). This framework emphasizes a shift in event focus
from the agent’s action to the patient’s change and the subsequent result. Although
scholars from diverse frameworks have investigated the *bǎ*-construction in terms of its
syntactic structure, semantic constraints, and information structure (Jing-Schmidt &
Tao, 2009; Liu, 2025; Peyraube, 1989; Sun, 1995; Yu et al., 2021; Zou, 1993; Zuo &
Li, 2025), these studies collectively demonstrate that its intricate internal form-meaning
relationships cannot be comprehensively captured by a single classification framework.

From the perspective of Cognitive Construction Grammar (CxG) (Croft, 2001;
Goldberg, 1995), the *bǎ*-construction is no longer regarded as a single syntactic pattern
but rather as a large and complex construction family. Within this family, two
fundamental organizational mechanisms operate simultaneously. On the one hand,

different sub-constructions are linked to higher-level schemas through inheritance relations, sharing an abstract causative event frame. On the other hand, constructions at the same level form horizontal links based on formal or semantic similarities, thereby constituting a multidimensional network (Liu, 2025; Song & Yang, 2022; Wang, 2020). At the semantic level, extensive corpus-based research indicates that *causation with displacement* constitutes the prototypical meaning of the *bǎ*-construction, wherein an agent causes a patient to undergo spatial displacement through an action (Lyu, 2015; Qian & Xiong, 2022; Zhang, 2001). Psycholinguistic studies further demonstrate that this displacement schema is automatically activated during comprehension. When processing *bǎ*-constructions, participants prioritize the construction of event representations involving spatial movement (Gao, 2004; Zhang et al., 2025). Furthermore, it is widely recognized that spatial displacement serves not only as the prototype meaning of the *bǎ*-construction but also as the starting point for its semantic extensions (Zhang, 2000, 2019). From a usage-based perspective, Bybee (2010) notes that constructional prototypes typically emerge from high-frequency members, as frequent exemplars best represent the category core, attract new members, and maintain semantic stability. Consequently, the high frequency of the causation-displacement construction in corpora makes it not only the semantic center of the *bǎ*-construction family but also the cognitive foundation for the semantic expansion of other sub-constructions.

Within the *bǎ*-construction family, the caused-motion and resultative constructions are two highly representative sub-constructions. They exhibit a relationship characterized by both inheritance and differentiation in terms of semantic focus and structural realization (Ziegeler, 2000). Formally, both sub-constructions share the basic syntactic frame (NP₁ + *bǎ* + NP₂ + V + XP), yet they differ significantly in the type of the post-verbal constituent. Typical structure of the caused-motion construction is NP₁ + *bǎ* + NP₂ + V + locative/path adjunct, whose semantic core lies in the agent causing the patient to undergo spatial displacement. In contrast, the resultative construction typically follows the pattern NP₁ + *bǎ* + NP₂ + V + resultative complement, focusing on the agent causing the patient to undergo a change of state.

- 105 (1) a. 小明 把 桌子 搬进 客厅。
 106 Xiaoming bǎ zhuōzi bān -jìn kètīng.
 107 Xiaoming BA table move-ALL living.room
 108 ‘Xiaoming moved the table into the living room.’ (‘X causes Y to move to Z’)
 109 b. 小王 把 门 打开了。
 110 Xiaowang bǎ mén dǎ -kāi -le
 111 Xiaowang BA door hit-open-PFV
 112 ‘Xiaowang opened the door.’ (‘X causes Y to become Z’)

113 In (1a), the path phrase *jìn kètīng* (进客厅 ‘into the living room’), positioned sentence-
 114 finally, marks the spatial endpoint of the patient. In (1b), the resultative complement
 115 *kāi-le* (开了 ‘open’) functions as a resultative complement, indicating the resultant
 116 state following the completion of the action. Beyond structural constraints, the two
 117 constructions also differ in their verb selection: caused-motion constructions favor
 118 verbs expressing direction or movement, whereas resultative constructions require
 119 verbs that can license a specific resultative outcome (Luo 2022; Lyu 2015; Qian &
 120 Xiong 2022). Thus, although the two constructions share an identical syntactic frame,
 121 they exhibit stable distinctions in the sentence-final constituents, justifying their
 122 treatment as two distinct sub-constructions within the *bǎ*-construction family.

123 Semantically, both the caused-motion and the resultative constructions inherit the
 124 prototypical semantics of the *bǎ*-construction (“X causes Y to achieve a certain result
 125 through an action”), yet they diverge in the dimension of result realization. The caused-
 126 motion construction manifests the patient’s displacement from a source to a goal within
 127 the spatial domain, whereas the resultative construction instantiates the transition from
 128 an initial state to a result state within the state domain. These two constructions are
 129 unified within the CAUSE–BECOME chain, corresponding respectively to the path
 130 realization and state realization of that chain (Boas, 2003). In cognitive terms, a change
 131 of location can be mapped onto state change, reflecting a path-based isomorphism
 132 between the two domains (Jackendoff 1983; Lakoff 1990). Accordingly, their
 133 relationship can be viewed as a semantic continuum extending from spatial causation
 134 to resultative causation. As demonstrated by Zuo & Li (2025) in a diachronic corpus

study, the event semantics of the *bǎ*-construction display a property of gradualness, suggesting that spatial causation and resultative causation are not dichotomous but continuously related, with the caused-motion construction providing the conceptual model for the emergence of the resultative construction.

However, while existing research has elucidated the similarities and divergent paths of these two constructions from formal and semantic perspectives, their mode of representation at the cognitive level remains unclear. In particular, there is a lack of direct psycholinguistic evidence regarding whether their relationship within the constructional network manifests as a hierarchical inheritance link or a horizontal linkage at the same level. This issue constitutes a key debate in current constructional network research and provides the theoretical motivation for employing the structural priming paradigm to investigate the psychological linkage between constructions.

2.2. Constructional Links under the Structural Priming Paradigm

As previously discussed, the caused-motion and resultative constructions exhibit both significant similarities and systematic divergences in their syntactic forms and event semantics. However, such descriptions at the formal and semantic levels are insufficient to explain how these two constructions are organized within the cognitive system. To fully reveal the linkage mechanisms between them within the *bǎ*-construction family, it is necessary to adopt a constructional network perspective to examine their hierarchical structures and functional connections in mental representations, and to clarify how these similarities are realized cognitively.

Under the framework of Construction Grammar, two primary theoretical approaches have been proposed regarding the relationship between the caused-motion and resultative constructions. One perspective emphasizes vertical inheritance between constructions. Goldberg (1995) explains the relationship through a metaphorical inheritance link, grounded in the conceptual resemblance between spatial motion and change of state, thus establishing the resultative as a metaphorical extension of the

caused-motion construction. This relationship reflects the hierarchical nature of the constructional network, in which the resultative construction, as a subordinate member, inherits the causative structure of the caused-motion construction but shifts its semantic focus toward the domain of results. In contrast, another perspective argues that a purely vertical inheritance relationship is insufficient to fully explain the links between these two target constructions (Diessel, 2019, 2023; Goldberg & Jackendoff, 2004; Ungerer, 2021). Goldberg & Jackendoff (2004) categorize them as two subtypes, the path construction and the property construction, arguing that they share a common superordinate schema but diverge into distinct branches in terms of specific semantic realization. This type of parallel differentiation exemplifies the horizontal expansion mechanism of constructional networks. Diessel (2019, 2023) further contends that linguistic networks are not purely hierarchical systems but essentially multidimensional structures sustained by both vertical inheritance and horizontal association. The former captures how the resultative construction inherits the causal-event frame of the caused-motion construction, while the latter reflects the associative links established between neighboring constructions due to formal or functional similarity. Applying these theoretical frameworks to the Mandarin *bǎ*-construction family, it is plausible that the caused-motion and resultative constructions may simultaneously manifest characteristics of both hierarchy and coordinate parallelism.

Building on this, structural priming research provides an operable empirical manner of distinguishing between different types of constructional relations. The underlying logic is that structural priming reflects spreading activation within a network of constructions (Diessel, 2019), and that different types of constructional relations may give rise to different patterns of activation spread (Ungerer, 2021, 2023b). Existing studies, using a self-paced reading paradigm with maze-task, have demonstrated that distinct priming patterns can, to a certain extent, reflect the psychological links between constructions. For instance, Ungerer (2021: 410-411) observed mutual inhibitory priming between English caused-motion and resultative constructions; because the effects were of comparable magnitude in both directions, he interpreted them as evidence for a bidirectional horizontal link between two constructions that are stored

as distinct but related representations. Similarly, Hong and Ma (2024: 388–391), investigating L2 English learners through a structural priming paradigm, found that while bidirectional inhibitory priming between these two constructions aligns with horizontal links, a unidirectional effect from the caused-motion to the into-causative construction supports an inheritance link. However, this correspondence is not absolute. Bouso et al. (2024: 335–337), examining priming between closely related constructions at the same level of abstraction, demonstrated that even sister constructions can exhibit significantly unidirectional priming patterns: the Superlative Objoid primed the Cognate Objoid across two experiments, while the reverse was not significant. This suggests that unidirectional priming does not necessarily point to a superordinate-subordinate inheritance relationship, but may instead stem from differences such as frequency, markedness, and variations in activation accessibility. Building on this, we propose that unidirectional priming need not always signal a superordinate–subordinate inheritance relationship; it may also reflect asymmetrical horizontal links in which more accessible, more entrenched constructions spread activation to semantically adjacent members more readily. This perspective offers a novel explanatory lens for re-evaluating the unidirectional inhibitory priming between the caused-motion and resultative constructions in Mandarin.

Within the Mandarin *bǎ*-construction family, the caused-motion and resultative constructions are likely to manifest both types of connections simultaneously. While previous research generally confirms their structural and semantic similarities as well as divergences, two key questions remain unresolved: First, should the relationship between the two constructions be understood as unidirectional semantic inheritance, or as a bidirectional functional association? Second, at the level of psychological representation, are they stored and activated as independent yet competing entities? These two questions constitute the central issues in current research on constructional networks and provide the theoretical basis for employing structural priming experiments to empirically examine the cognitive linkage between constructions.

2.3. Structural Priming in Constructional Network

The linkage between constructions often relies on the complex interaction between form and function. To employ appropriate research methods to detect the structural and semantic interactions between constructions, one must remain sensitive to both syntactic and semantic levels. The structural priming paradigm precisely satisfies this need. By examining how sentence processing mutually influences subsequent structural choices, the structural priming paradigm reveals shared mental representations and activation mechanisms across linguistic structures (Bock, 1986; Pickering & Branigan, 1998). Early studies suggested that structural priming effects were solely triggered by syntactic repetition (Bock & Loebell, 1990). However, subsequent research recognized that priming can also arise from similarities in semantic roles, event structure, or information structure (Bunger et al., 2013; Hare & Goldberg, 1999; Ziegler & Snedeker, 2018). This indicates that structural priming is highly sensitive to form-function mappings and thus serves as one of the most effective empirical tools for testing the psychological reality of constructional networks.

In previous research, the structural priming paradigm has predominantly focused on alternating constructions, such as active-passive alternation or dative alternations (Mahowald et al., 2016). These constructions can be regarded as formal variants of a shared underlying meaning, or *allostructions* (Cappelle, 2006), which makes them particularly suitable for comparison of syntactic choice in language production experiments. However, non-alternating constructions are typically non-interchangeable, making them difficult to investigate through traditional production tasks. To address this issue, Ungerer (2021) pioneered the extension of structural priming to non-alternating constructions using the maze task. By controlling the reading path word-by-word and requiring participants to choose between a target word and a distractor at each step, this method reflects processing load and structural expectations in real-time during comprehension. The study found that even semantically non-equivalent constructions can mutually influence each other during comprehension. Subsequent research by Bouso et al. (2024) further validated this paradigm in comprehension tasks, demonstrating that structural priming can reveal sisterhood relations between constructions, specifically horizontal links based on shared semantic frameworks. This

shift signifies that structural priming has transcended syntactic similarity to become a psycholinguistic method capable of exploring the organization of semantic networks within constructions. Compared with traditional production paradigms, comprehension-based priming tasks do not rely on alternation between constructions but instead directly reflect the processing load and activation patterns involved in language understanding. Consequently, structural priming experiments exhibit strong usage-based orientation, with participants' comprehension processes reflecting the spreading activation of constructions across linguistic experience (Traugott & Trousdale, 2013).

In exploring priming mechanisms, a pivotal question concerns whether priming effects reflect abstract constructional representations or are merely driven by surface-level lexical overlap. Verb repetition, as a critical variable in structural priming research, is commonly employed to distinguish between lexically driven and abstract constructional priming (Pickering & Branigan, 1998). Traditional accounts indicate that priming effects are significantly enhanced when the prime and target share the same verb, a phenomenon termed the lexical boost effect, reflecting strengthened associations between verbs and constructions. However, evidence from comprehension tasks show divergent patterns. Some studies have shown that priming effects depend critically on verb overlap (Arai et al., 2007; Branigan et al., 2005), while others have demonstrated significant priming even in the absence of lexical repetition (Fine & Jaeger, 2016; Kim et al., 2013; Thothathiri & Snedeker, 2008). Ungerer (2021) further tested this issue and found that verb repetition produced a limited facilitation effect during verb processing, but had only a limited impact on the overall cross-construction priming pattern, reaching marginal significance in only a few conditions. Crucially, these effects were both condition-specific and temporally localized, emerging only in restricted contexts and not persisting across subsequent processing stages. Taken together, these findings suggest that the role of verb overlap is primarily confined to specific temporal windows rather than systematically altering the priming relationships between constructions. Therefore, rather than determining constructional relations, verb repetition merely modulates activation intensity during short-term processing.

In summary, the structural priming paradigm demonstrates both formal and semantic sensitivity in methodology, enabling it to reveal shared features, competitive relationships, and hierarchical organization among constructions at the processing level. Its comprehension-based implementation offers a novel empirical approach to examining non-alternating constructional relations, providing a powerful tool for testing the psychological organization of constructional networks. Given that the caused-motion and resultative constructions within the Mandarin *bǎ*-construction family exhibit formal similarity but semantic divergence, the present study adopts the experimental framework proposed by Ungerer (2021, 2022, 2023a, 2023b). Specifically, a maze self-paced reading task is employed to investigate the mutual influences between the two constructions and the role of verb repetition, with the aim of revealing their connection mechanisms and hierarchical properties in mental representation.

3. Experiment

3.1. Research Question and Prediction

Following the research framework of Ungerer (2021, 2022, 2023b), the present study utilizes the structural priming paradigm to investigate the psychological association between the caused-motion construction and the resultative construction in Mandarin. Drawing on the analytical logic of prior studies that infer constructional network link types by comparing the directionality and magnitude of priming effects between non-alternating constructions (Ungerer, 2021), three specific research questions are proposed:

- RQ1: Does a cross-construction priming effect exist between the caused-motion and resultative constructions? If so, does this effect manifest as facilitation or inhibition in processing?

The present study posits that these two constructions share a highly similar surface framework at the syntactic level and the same prototypical semantic frame at the semantic level. This combination of formal similarity and semantic overlap provides the basis for the mutual activation of these two constructions in mental representation. Therefore, cross-construction priming effects are expected to occur during comprehension. Given that both types of constructions in the Mandarin *bǎ*-sentence have high usage frequencies and robust representations, a significant yet relatively weak priming effect is predicted. A facilitatory effect would suggest a shared activation base in mental representation, whereas an inhibitory effect might reflect a competitive relationship between the two constructions.

— RQ2: Does the cross-construction priming effect exhibit directionality? Specifically, can the caused-motion construction prime the resultative construction, and does the reverse priming also occur?

As discussed earlier (see Section 2.1), the meaning of spatial displacement constitutes not only the prototypical semantics of the *bǎ*-construction but also the starting point of its semantic extension (Gao, 2004; Lyu, 2015; Qian & Xiong, 2022; Zhang, 2001; Zhang, 2000, 2019; Zhang et al., 2025). Accordingly, the present study expects the priming effect to exhibit asymmetric characteristics. As the prototypical construction, the caused-motion construction possesses higher activation accessibility and is expected to significantly prime the resultative construction through spreading activation. Conversely, as a semantic extension that inherits its event frame from the caused-motion construction, the resultative construction is expected to produce a weak or non-significant reverse priming effect on the prototypical construction.

— RQ3: Does verb repetition modulate the cross-construction priming effect? If so, does lexical overlap have a consistent influence on the priming patterns between the two constructions?

It is hypothesized that verb repetition will have only a limited influence on the overall

pattern of cross-construction priming. Since the maze reading task primarily captures participants' event integration processes rather than syntactic choice, lexical overlap is expected to produce only a limited facilitation during verb processing, without substantially strengthening the construction-level priming effect. Therefore, if significant cross-construction priming is still observed under the non-repetition condition, it would further support the view that constructions are represented as lexically independent abstract schemas. In contrast, if the priming effect occurs only under the verb-repetition condition, it would suggest that the effect relies more heavily on lexical-level associations.

Taken together, by systematically investigating the presence, directionality, and lexical modulation of cross-construction priming, this study aims to reveal the psychological linkage patterns between the caused-motion and resultative constructions at the processing level. These findings will serve as the basis for inferring their organizational structure within the Mandarin *bǎ*-constructional network.

3.2. Participants

In the present study, 160 native Mandarin speakers were recruited through offline convenience sampling. All participants, with ages ranging from 18 to 45 years old, reported no history of language disorders or cognitive impairments, and provided informed consent prior to participation. In addition, following the criterion of Boyce and Levy (2023), and given the high cognitive demands of the maze task, the sample was restricted to participants holding a bachelor's degree or higher. This restriction on educational qualifications was implemented to minimize the likelihood of random responding and ensure the reliability of the behavioral data. Table 1 presents the demographic characteristics of the participants.

Table 1: Participant background information.

Metric	Measure	Value	Percentage
Age	Average	25.3	-
	Median	24	-

	Mode	23	-
Gender	Female	92	57.50%
	Male	68	42.50%
	PhD	13	8.13%
Education	Postgraduate	94	58.75%
	Undergraduate	53	33.13%
Handedness	Left	9	5.63
	Right	151	94.38

3.3. Experimental Design and Materials

Following Ungerer (2021), the present study employed the maze variant of the self-paced reading task (Forster et al., 2009) to ensure participants' deep integration of contextual information and to minimize spillover effects (Boyce et al., 2020). Previous research has shown that structural priming effects reflect shared mental representations and the spreading activation between different linguistic structures (Bock, 1986; Pickering & Branigan, 1998). Moreover, comprehension-based paradigms such as the maze task are highly sensitive to capturing shifts in structural expectations during sentence comprehension (Bouso et al. 2024; Ungerer 2021). Leveraging these methodological advantages, the present study systematically examines the cross-constructional priming effects that arise when native speakers comprehend and process *bă*-constructions instantiating the caused-motion and resultative constructions, thereby providing empirical evidence to reveal the representational links and cognitive mechanisms underlying these two construction types.

In the maze task, each target word was presented simultaneously with a distractor word, and participants were required to select the word that could plausibly continue the sentence. The distractors were generated using the Multilingual A-maze program developed by Levinson et al. (2023). This program, based on the Bidirectional Encoder Representations from Transformers (BERT) language model (Devlin et al., 2019), selected candidate distractor words by evaluating multiple dimensions such as word

frequency and syntactic consistency. All automatically generated distractors were subsequently inspected and refined by human reviewers to ensure that none could combine with the preceding words to form a grammatically correct sentence.

The experimental materials focused on two non-alternating constructions within the Mandarin *bǎ*-construction family. Sixteen high-frequency verbs (e.g., 放 ‘put’, 推 ‘push’, 拉 ‘pull’, 扔 ‘throw’, etc.) were selected, and for each verb, two caused-motion sentences and two resultative sentences were created, yielding a total of 64 critical sentences. The caused-motion construction was structured as “NP₁ + *bǎ* + NP₂ + VP + locative phrase,” whereas the resultative construction followed “NP₁ + *bǎ* + NP₂ + VP + resultative complement.” To control for potential non-structural interference, all subject nouns (NP₁) were replaced with de-identified names (e.g., Xiao Li, Xiao Shi) to eliminate socio-demographic factors during the sentence processing. None of the names were repeated across stimuli. Examples of experimental materials are provided in Figure 1, and the complete list of stimuli can be seen in Appendix A.

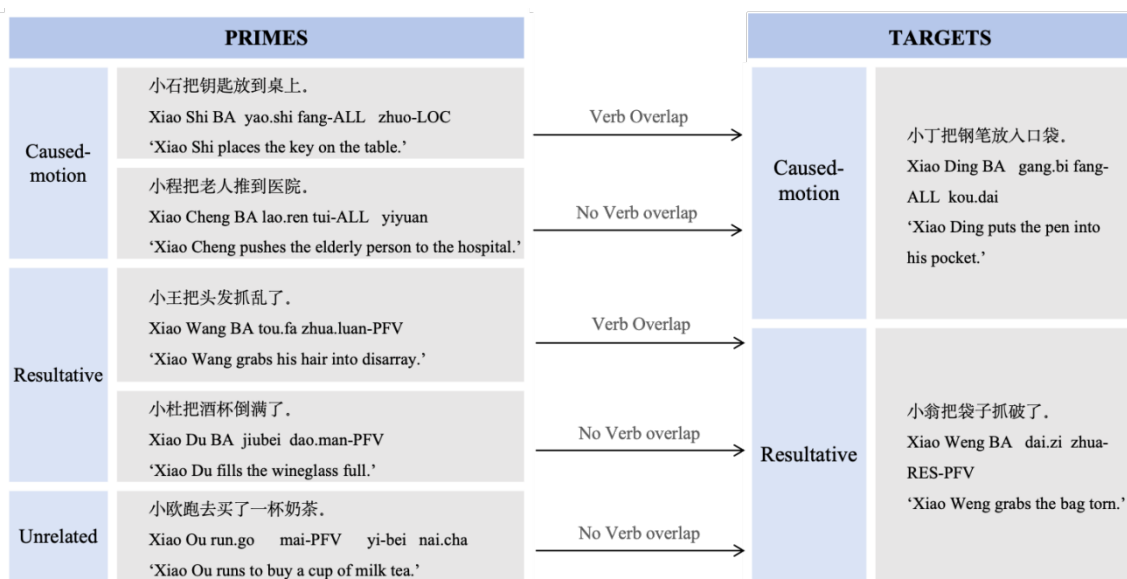


Figure 1: Example of experimental primes and targets with and without verb overlap.

Two rounds of pre-tests were conducted to examine all critical sentences prior to the formal experiment. Twenty native speakers, served as independent raters, evaluated the acceptability of each sentence by using a 7-point Likert scale (1 = completely unacceptable, 7 = completely natural and acceptable). Only sentences with a median

rating of 6 or higher were retained for the experiment.

In addition to the 64 critical sentences, 96 filler sentences were also included in the materials. These fillers were structurally and semantically unrelated to the critical items and covered a variety of sentence types, including verb-object, verb-complement, adverb-verb, adverb-verb-complement, serial-verb, pivot, adverb-adjective, and adjective-complement constructions. Following Ungerer's (2021) paradigm, a subset of the fillers served as "unrelated" or "no-prime" items providing materials for the unrelated baseline condition, whereas the remaining fillers functioned as distractors to prevent participants from detecting the experimental purpose.

The experiment adopted a pseudo-randomized nested design with a 3 (prime construction type: CM "caused-motion" / RES "resultative" / UNREL "unrelated") $\times$ 2 (target construction type: CM "caused-motion" / RES "resultative") factorial structure. Based on this design, the 64 critical sentences were paired to form 32 prime–target combinations distributed across three experimental conditions: (i) within-construction priming (CM $\rightarrow$ CM and RES $\rightarrow$ RES), which tested whether prior exposure to a construction facilitates subsequent processing of the same construction; (ii) cross-construction priming (CM $\rightarrow$ RES and RES $\rightarrow$ CM), which examined whether processing one construction influences the comprehension of the other; and (iii) an unrelated baseline condition (UNREL $\rightarrow$ CM/RES), which served as a control for evaluating priming effects. Within-construction priming effects were assessed by comparing within-construction pairs with the baseline condition, while cross-constructional priming was evaluated by comparing cross-construction pairs against the same baseline. Eight counterbalanced stimulus lists were constructed, each comprising 160 sentences. Each list was divided into four blocks with eight randomly ordered prime-target pairs per block. Furthermore, the materials were arranged in line with Ungerer's (2021) pseudo-randomized interspersed design to control for structural variables. All prime-target pairs, filler sentences, and baseline items were intermixed, with two to four unrelated fillers inserted between any two critical pairs to prevent short-term structural repetition effects.

3.4. Division of Critical Regions

As illustrated in Figure 2, the present study focuses on two critical regions of interest: the verb, and the first word of the complement. The identification of these regions is grounded in the syntactic and semantic properties of the *bǎ*-construction and the functional characteristics of syntactic processing, where each position serves distinct information-integration roles during incremental sentence comprehension. Consequently, these regions exhibit differential sensitivity to structural and lexical priming effects.

				Control region for Verb overlap: Verb	Control Region: First word of Complement (Preposition for caused-motion; Adjective for resultative)	
	小丹	把	钥匙	放	到	桌上
Caused-motion	Xiaodan	BA	key	put	to-ALL	table-LOC.
(Xiaodan put the key on the table.)						
	小兴	把	窗户	推	开了	
Resultative	Xiaoxing	BA	window	push	open-PFV	
(Xiaoxing pushed the window open.)						

Figure 2: Regions of interest for the caused-motion and resultative constructions.

The first word of the complement marks the point at which the caused-motion and resultative constructions fully diverge. In the caused-motion construction, this element, typically a preposition, encodes the spatial direction and endpoint of movement, forming the central semantic component of the caused-motion event. In contrast, in the resultative construction, the complement directly specifies the resultant state. As this complement carries the most recognizable semantic features of the construction, it serves as the critical node for late-stage integrative processing and is the position most likely to exhibit structural priming effects.

In contrast, the verb is not treated as a primary critical region for constructional differentiation in the present experimental design. Since both constructions share the same verb set, the verb itself provides no evidence as to whether the unfolding sentence

instantiates a caused-motion or a resultative pattern. However, during real-time processing, participants may already begin integrating the semantic information of the object (NP₂) and the verb at this position, forming preliminary expectations about the upcoming constructional type. Nonetheless, compared to the complement first word region, such predictions based on the verb–object combination remain markedly uncertain and do not yield sufficient information to stably determine the construction type. Given this uncertainty, the present study does not regard the verb slot as a critical site for testing constructional priming effects. Instead, it is treated as an ideal location for detecting the lexical boost effect (Ungerer, 2021). For this reason, the present study only observed the main effect of verb repetition in this region and did not test interactions with construction type.

3.5. Procedure

The experiment was conducted in a laboratory setting. After receiving a brief introduction to the task and signing the informed consent form, participants completed the experiment via the PsychoPy platform (Peirce et al., 2019). Once the participant’s demographic information was entered, the system automatically presented detailed on-screen instructions outlining the experimental procedure and task requirements. Before the formal experimental session, each participant completed nine practice trials to become familiar with the task flow and response procedure.

At the beginning of each trial, a fixation cross (“+”) appeared at the center of the screen for 2000 milliseconds before disappearing. In each experimental trial, participants read the sentence word by word using a two-alternative forced-choice procedure, choosing between the target word and a distractor word displayed side by side at the center of the screen. The left-right position of the two words was randomized. Participants pressed the “W” key to select the word on the left and the “I” key to select the one on the right. If an incorrect word was chosen, the system immediately displayed an error message and terminated that trial. To ensure that participants selected the correct sentence onset, the first word of each sentence was primed with the pseudo-

word distractor “X-X-X”. The system precisely recorded the participants' reaction time data at each word position.

The experiment included scheduled breaks after approximately 25%, 50%, and 75% of the total trials. Each break lasted one minute, and at least two filler trials followed each break to help participants reestablish their reading rhythm. The entire session lasted approximately 40–45 minutes. The presentation format of the maze task is illustrated in Figure 3.

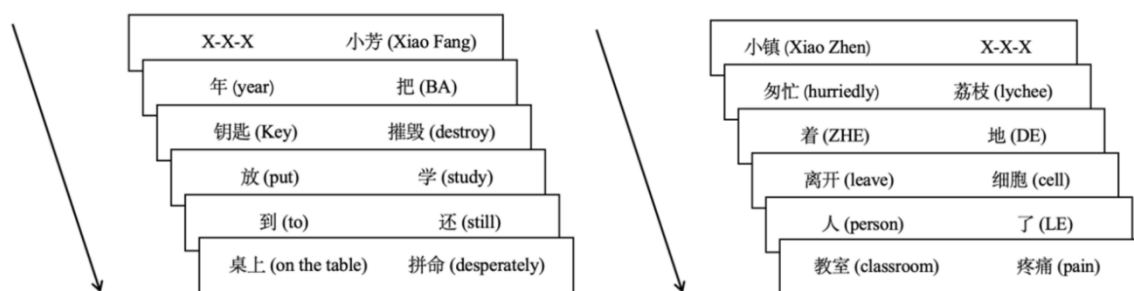


Figure 3: A sample of the maze task

3.6. Data analysis

The reaction-time data were analyzed using linear mixed-effects models, implemented in R (R Core Team, 2019). This study focused exclusively on reaction time data for statistical analysis. Although accuracy rates were recorded during preprocessing and served as strict criteria for trial and participant exclusion, they were not incorporated into subsequent modeling. This methodological choice stems from the nature of the maze self-paced reading task, wherein participants could achieve high accuracy by inference even when encountering processing difficulties with target sentences, thereby limiting the statistical power of accuracy rates in detecting subtle processing differences across experimental conditions (Bouso et al., 2024). Consequently, in line with common practice in this paradigm, systematic statistical comparisons and reporting of accuracy rates were not performed in this study.

Prior to statistical modeling, the dataset underwent systematic cleaning and

preprocessing. First, to ensure the materials met the controlled requirements of the structural priming paradigm (Ungerer, 2021) and to eliminate potential interference from complement form repetition or verbal semantic differences, a small number of confounding trials were pre-emptively excluded to enhance the explanatory validity of the results. Subsequently, participants whose error rate on target sentences exceeded 50% were excluded. Trials with incorrect judgments were removed, as were extreme responses shorter than 200 ms or longer than 3000 ms. In addition, outliers were trimmed on a per-participant basis using a ± 2.5 SD criterion. As raw reaction times exhibited a right-skewed distribution, all RTs were log-transformed (base 10) to meet the normality assumptions of linear modeling (Baayen & Milin, 2010). The final dataset contained 22,714 valid trials.

During statistical analysis, linear mixed-effects modeling was used to capture how relations between constructions manifest dynamically during real-time processing. Models were fitted using lme4 (Bates et al., 2015) and lmerTest (Kuznetsova et al., 2017). The primary analyses focused on the complement-first region, evaluating how different prime constructions influenced the processing of target constructions. Fixed effects included target construction (caused-motion vs. resultative), prime construction (unrelated vs. caused-motion vs. resultative), and their interaction, with participant age entered as a covariate. The random-effects structure included random intercepts for participants, target sentences, and prime sentences, as well as by-participant slopes for target construction to capture systematic individual differences in construction processing.

To examine the role of lexical overlap in cross-construction priming, verb repetition was incorporated as an additional factor. Given that unrelated primes never shared verbs with targets, the analysis of verb repetition was restricted to trials where the prime was either a caused-motion or a resultative construction. As in the primary model, analyses were conducted for the complement- first region. The model contained a three-way interaction among *prime construction*, *target construction*, and *verb repetition*, which determines whether cross-construction priming is modulated by shared lexical material and to evaluate the functional role of verb repetition within the

532 constructional network.

533 Finally, the verb region was analyzed independently. As discussed in Section 3.3,
534 the verb region does not involve construction-specific semantic-role mapping or
535 syntactic integration. Therefore, prime construction was not included as a predictor for
536 this region. The model included only verb repetition (repeated vs. non-repeated) as a
537 fixed effect, and thereby determine whether lexical repetition facilitates processing at
538 the level of specific lexical access. By modeling this region separately, the study
539 distinguished pure lexical repetition effects from structural priming effects arising
540 during syntactic integration, thereby clarifying the distinct functional contributions of
541 lexical items and constructions in online processing.

542 Post-hoc comparisons for all critical effects were conducted using emmeans
543 (Lenth et al., 2025), with p-values adjusted using Tukey correction. While the statistical
544 models included main effects and interactions of prime construction, target construction,
545 and verb repetition, the primary analytical focus of this study lies in condition-specific
546 contrasts between prime–target combinations, rather than in the omnibus model terms
547 themselves. This approach follows established practice in structural priming research,
548 where interaction effects are often interpreted through pairwise comparisons that
549 capture localized facilitation or inhibition patterns (Bouso et al., 2024; Ungerer, 2021).
550 Accordingly, only fixed effects directly relevant to the research questions—namely,
551 condition-specific contrasts—are reported. Main effects and covariates (e.g., age) were
552 included as controls but are not interpreted. Random effects were included to account
553 for variance across participants and items and are not discussed further.

555 4. Results

556 4.1. Priming effects between Caused-Motion and Resultative 557 Constructions

558 Linear mixed-effects models were constructed to investigate the influence of

prime and target constructions on reaction times at the complement-first slot. The omnibus interaction between target and prime constructions did not reach statistical significance [$F(2, 57.79) = 0.91, p = 0.408$], with small effect sizes ($target_cxn\ RES \times prime_cxn\ RES: \beta = -0.026, SE = 0.046$; $target_cxn\ RES \times prime_cxn\ UNREL: \beta = -0.054, SE = 0.041$). In this region, a main effect of prime construction reached significance [$F(2, 57.79) = 3.20, p = 0.048$]. Regarding effect sizes, the models accounted for a relatively small proportion of variance at the fixed-effect level ($R_m^2 = 0.031$), whereas the explanatory power increased after incorporating random effects ($R_c^2 = 0.324$).

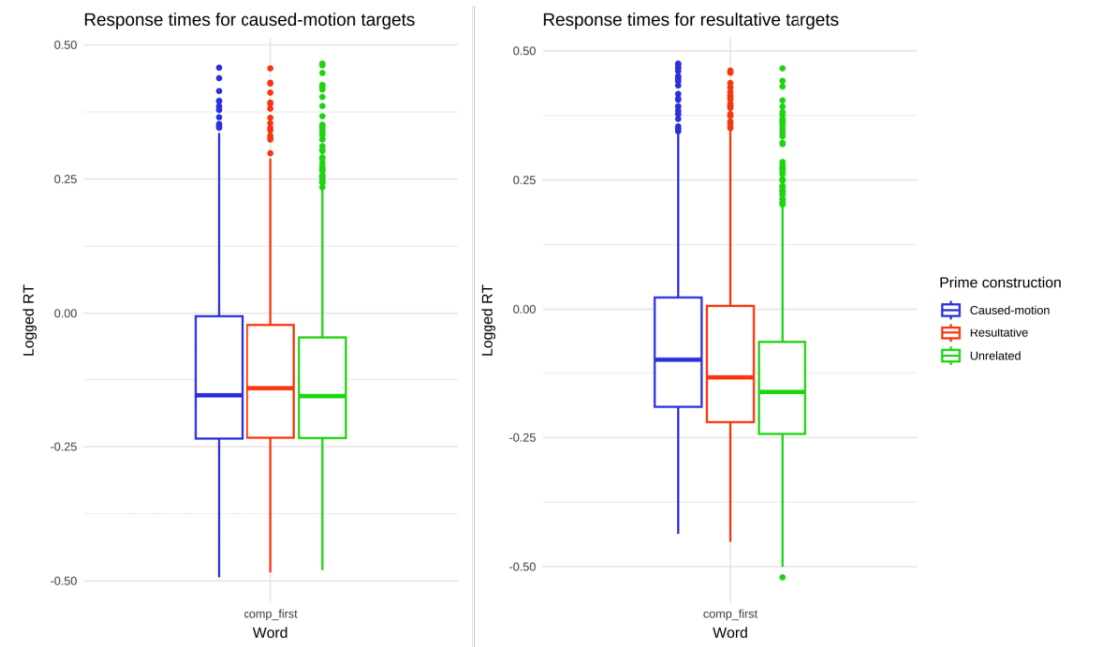


Figure 4: Log-transformed reaction times for caused-motion and resultative sentences under different prime conditions

Despite the absence of a significant omnibus interaction, Tukey-adjusted post-hoc comparisons were conducted to further examine condition-specific differences between prime–target combinations. This approach is widely adopted in structural priming research, where potential facilitation or inhibition effects may emerge in specific contrasts rather than as a uniform interaction across all conditions (Bouso et al., 2024; Ungerer, 2021). Table 2 presents the post-hoc comparisons among the three prime conditions (CM, RES, and UNREL) for each target construction (CM and RES), for

the complement-first positions. The two rightmost columns of the table present the effect sizes for all contrasts, with differences on the log-transformed scale back-transformed into absolute reaction-time differences (ms) and relative percentage changes. Figure 4 provides a visual illustration of the reaction-time patterns across conditions.

Table 2: Effects of different prime constructions on target sentence processing: reaction-time comparisons at the complement-first region.

Target cxn	Diff. Between prime cxns (A minus B)		Estimate	SE	Z	P	Sign.	Diff.in ms	Relative diff.
Critical Region: First word of complement phrase (preposition in CM, adjective in RES)									
CM	CM-RES	0.0044	0.0344	0.127	0.9911				
	CM-UNREL	0.0217	0.0302	0.718	0.7525				
	RES-UNREL	0.0173	0.0287	0.605	0.8176				
RES	CM-RES	0.0301	0.0309	0.973	0.5941				
	CM-UNREL	0.0752	0.0271	2.775	0.0153	*	136.18	+15.90%	
	RES-UNREL	0.0451	0.0271	1.664	0.2191				

CM = caused-motion; RES = resultative; UNREL = unrelated; '*' p < 0.05, '**' p < 0.01; '***' p < 0.001; Results with p < 0.05 in bold.

Priming effects at the complement-first region are shown in Figure 4 and Table 2. For caused-motion targets, reaction-time differences among the three prime conditions were not statistically significant, indicating that processing at this position was largely insensitive to prime construction type. In contrast, for resultative targets, prime construction type had a significant effect at this slot. Specifically, under caused-motion prime conditions, reaction times at the complement-first slot were significantly longer than under unrelated prime conditions, with a difference of 136.18ms (representing a relative slowdown of 15.90%), revealing a clear inhibitory priming effect. Although caused-motion primes also tended to yield slower responses than resultative primes, this contrast did not reach statistical significance.

4.2. Verb repetition effects

At the complement-first slot, verb repetition exhibited a significant interactive effect with constructional factors. Specifically, the three-way interaction among verb repetition, prime construction, and target construction reached statistical significance ($\beta = -0.255$, $SE = 0.100$, $t = -2.545$, $p = 0.016$), indicating that the impact of verb repetition on reaction times varies with the specific prime-target construction pairing. Post-hoc comparisons revealed that, in the caused-motion prime – caused-motion target condition, verb overlap significantly facilitated processing, with reaction times shortened by approximately 193ms (a relative acceleration of 22.7%) compared to the no-overlap condition. Conversely, in the caused-motion prime – resultative target condition, verb overlap showed an inhibitory trend, with reaction times increasing by approximately 167.1ms (a relative deceleration of 21.5%) compared to the no-overlap condition, although this contrast was marginally significant. No statistically significant verb repetition effects were observed in the remaining constructional combinations.

Table 3: Impact of Verb Overlap on Processing Construction Pairings at Critical Regions

Target Cxn	Prime Condition	Estimate	SE	Z	P	Sign.	Diff.in ms	Relative diff.
Critical Region: First word of complement phrase (preposition in CM, adjective in RES)								
CM	CM (Overlap vs No)	-0.1118	0.0568	-1.967	0.0492	*	-193	-22.70%
	RES (Overlap vs No)	0.0161	0.0501	0.321	0.7481			
RES	CM (Overlap vs No)	0.0846	0.0464	1.823	0.0683	.	167.1	21.50%
	RES (Overlap vs No)	-0.0426	0.0464	-0.919	0.3584			
Critical Region of Verb								
—	Overlap vs. No	-0.0261	0.0162	-1.609	0.118			

CM = caused-motion; RES = resultative; UNREL = unrelated; ‘.’ $p < 0.1$; ‘*’ $p < 0.05$; Results with $p < 0.1$ in bold; At the verb position, only the main effect of verb overlap was tested, without distinguishing between target/prime construction combinations.

At the verb position itself, verb repetition showed a marginally significant trend

toward a main effect ($\beta = -0.026$, $SE = 0.016$, $t = -1.609$, $p = 0.118$). Under the verb overlap condition, the predicted reaction time was approximately 802.5ms, whereas it was approximately 852.3ms in the no-overlap condition, representing a difference of about 49.8ms (a relative change of approximately 6.2%).

5. Discussion

5.1. Existence of Cross-Constructional Priming

This study examined the structural priming effects between the caused-motion construction and the resultative construction within *bǎ*-construction during sentence comprehension. The results revealed a significant cross-construction priming effect between the two constructions. Specifically, in the critical region of complement first word, resultative targets were processed significantly slower following caused-motion primes, showing an inhibitory effect. By comparison, no significant differences were observed for caused-motion targets across priming conditions.

It should be noted that the present finding is consistent with the inhibitory cross-constructional priming reported by Ungerer (2021) in L1 English comprehension, in which exposure to a caused-motion prime slowed the processing of a subsequent resultative target relative to the unrelated baseline. In contrast to the bidirectional inhibitory priming reported between the two constructions by Ungerer (2021) in L1 and by Hong & Ma (2024) in L2 English, the present study uncovered a stable inhibitory pattern that operated unidirectionally—from the caused-motion prime to the resultative target—at the complement first word position within the Mandarin *bǎ*-construction. This pattern can be interpreted at multiple levels. Structurally, the complement slot constitutes a point of formal divergence between the two constructions. In the caused-motion construction, this position is typically introduced by a preposition to encode a spatial path and motion endpoint; However, in the resultative construction, the complement-first position is occupied by an adjectival component expressing a resultant state. Due to the fundamental differences in encoding at this position, when

the preceding prime activates the caused-motion construction, participants must adjust their existing structural expectations while processing the complement of the resultative target. This formal mismatch at the syntactic level constitutes a primary source of the observed inhibitory effect.

From a semantic perspective, the two constructions also differ in their organization of event structures. The caused-motion construction highlights an Agent-Causation-Path event frame, whereas the resultative construction focuses on Agent-Causation-State Change as its core semantics. Because the spatial-path schema activated by the prime mismatches the resultative semantics required by the complement in the resultative target, participants must suppress the pre-activated spatial-motion framework and reconstruct a new resultative event structure, leading to processing delays. This inhibitory effect, triggered by prediction violation and structural competition, aligns with explanations in priming research suggesting that unmet expectations increase processing costs (Jaeger & Snider, 2013; Kuperberg, 2007; Levy, 2008; Ungerer, 2021).

Furthermore, within the *bǎ*-construction family, this inhibitory effect is not accidental. As a higher-frequency and more prototypical member of the family (Lyu, 2015; Qian & Xiong, 2022; Zhang, 2001), the caused-motion construction's mental representation is more easily activated and spreads to semantically adjacent constructions (Bybee, 2010; Schmid, 2020). Consequently, when this highly activated event frame mismatches the semantic requirements of the resultative construction, it may amplify the competition between constructions, triggering a stronger inhibitory effect at the critical integration site.

Overall, this pattern suggests that speakers are sensitive to both the association and differentiation between the caused-motion and resultative during comprehension. While the two constructions share both a similar syntactic pattern (S + *bǎ* + O + V) and a fundamental causation-change event frame, their differentiation is reflected in the divergence of their semantic representations. These findings indicate that constructions encode basic event types, providing empirical support for the Scene Encoding Hypothesis proposed by Goldberg (1995). Moreover, the study corroborates the view

that priming effects reflect representational similarity (Pickering and Branigan, 1998), suggesting that cross-constructive priming is jointly constrained by structural configurations and constructive semantics. In line with Ungerer (2021), this pattern supports the claim that priming effects are rooted in the interaction of form and meaning at the level of constructive representations, rather than being driven by surface syntactic similarity alone.

5.2. Directionality of Priming and Prototype-Based Mechanism

This study further examined the directionality of cross-constructive priming within the Chinese *bǎ*-construction, that is, whether the priming between the caused-motion and resultative constructions manifests as bidirectional or unidirectional. Experimental results showed that priming effects were restricted to resultative targets, with no significant differences observed for caused-motion targets across different prime conditions. This indicates a unidirectional priming pattern where a caused-motion prime inhibited the subsequent processing of a resultative target, but no reciprocal effect was observed.

In terms of the mechanism of structural priming, priming effects are typically interpreted as a reflection of spreading activation within mental representations (Diessel, 2019). Within this framework, unidirectional priming does not directly equate to a predetermined constructive network relation but rather reflects an asymmetry in activation accessibility and similarity distribution between different representations. Specifically, the findings suggest that the caused-motion construction is more readily activated during processing and can exert a stronger priming influence on the semantically adjacent resultative construction, whereas reverse activation does not reach significance.

From the perspective of the *bǎ*-construction network, this unidirectional pattern is not accidental but reflects the positional and representational strength differences among constructions within the network. As a member exhibiting higher prototypicality and usage frequency within the network (Gao, 2004; Lyu, 2015; Qian & Xiong, 2022;

Zhang, 2001; Zhang, 2000, 2019; Zhang et al., 2025), the caused-motion construction demonstrates a higher degree of entrenchment in mental representation. Usage-based approaches have consistently shown that higher frequency and prototypicality are associated with greater entrenchment and increased resting activation levels (Bybee, 2010). Against this background, the observed priming asymmetry can be interpreted as reflecting a processing advantage of the caused-motion construction, which enables it to more readily spread activation toward semantically adjacent resultative constructions. Langacker (2008) argues that constructional systems typically exhibit a center-periphery organization, in which prototypical members occupy the semantic and structural center, while non-prototypical members are linked to them through processes of extension or variation. The unidirectional priming effect observed in the present study can thus be interpreted as a path of spreading activation from a highly prototypical member to a less prototypical one.

However, this asymmetry does not reflect a traditional superordinate-subordinate inheritance relation, but rather stems from differences in prototypicality and representational weight among constructions at the same hierarchical level. Crucially, both the caused-motion and resultative constructions in the present study are instantiations of the higher-level *bǎ*-construction schema, and thus already share a common superordinate node. Moreover, as the specific semantics of state-change is not predictable from spatial-motion, both require independent specification, leading Goldberg & Jackendoff (2004) to model them as parallel subtypes rather than an inheritance pair. As such, their relationship is structurally defined as horizontal within the constructional network, rather than vertical between different levels of abstraction. Unlike Ungerer (2021) and Hong & Ma (2024), who take bidirectional priming between English caused-motion and resultative constructions as evidence for horizontal links, the present study shows that the absence of bidirectionality does not necessarily imply vertical inheritance. If the observed priming asymmetry were driven by vertical inheritance, it should reflect activation flowing from the abstract *bǎ*-schema downward to both of its subordinate instances, yielding a pattern of generalized facilitation across construction types. However, the present findings instead show a selective and

directional effect confined to specific construction pairings, which is not characteristic of schema-to-instance generalization, where facilitation is expected to generalize across all subordinate constructions rather than being restricted to a single directional pairing. Supporting this view, Bouso et al. (2024) demonstrate that priming between constructions at the same level of abstraction can exhibit significant directionality, reflecting differences in accessibility and psychological distance rather than derivation or inheritance. Consequently, the unidirectional effect in this study is more plausibly explained as asymmetric, similarity-driven activation between horizontal sister constructions, rather than as evidence of vertical inheritance.

In addition, the observed directionality aligns with the concept of multidimensional constructional networks. Diessel (2023) emphasizes that vertical and horizontal links are not opposing mechanisms but complementary modes of linguistic organization. Vertical networks capture abstraction and generalization, revealing the overarching schemas shared across constructions, whereas horizontal links reflect connections among constructions at the same level that arise from similarities in form, meaning, or function. Together, they constitute a multidimensional network. As illustrated in Figure 5, the *bã*-construction network proposed here embodies this multidimensional architecture. On the vertical dimension, the caused-motion and resultative constructions are both instantiations of the higher-level *bã*-construction schema. Meanwhile, on the horizontal dimension, they exhibit an asymmetric psychological distance due to differences in the concreteness of the result. In contrast to previous studies reporting bidirectional priming between these constructions (Hong & Ma, 2024; Ungerer, 2021), the unidirectional effect here suggests that horizontal links in specific linguistic systems may manifest as asymmetric structures. Specifically, the caused-motion construction occupies a central position due to its higher prototypicality and semantic prominence, allowing its activation to spread horizontally to resultative construction, while reverse activation is constrained by the asymmetry in cognitive accessibility.

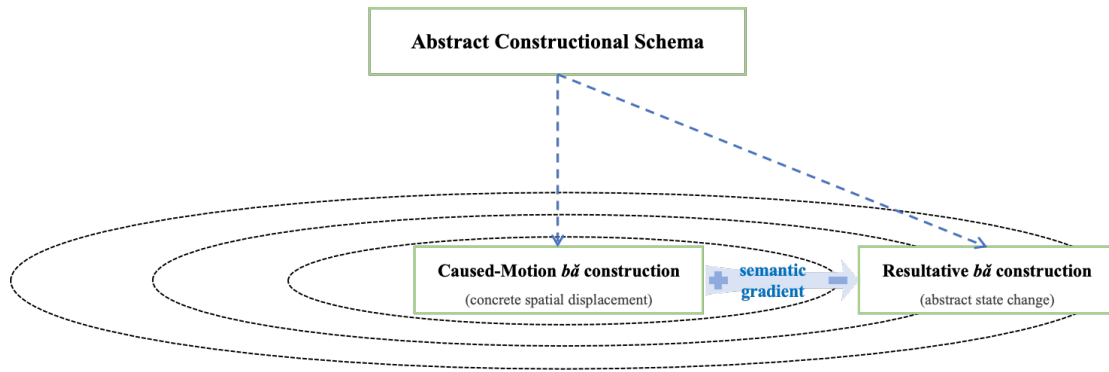


Figure 5: The Multidimensional Network of the *bǎ*-Construction

Furthermore, the directionality of this asymmetry reflects the semantic relationship within the event structures. Although both constructions share an abstract CAUSE–RESULT schema, they differ in their event-structural focus. The caused-motion construction emphasizes spatial trajectory and motion process to present the result, whereas the resultative construction highlights the endpoint of state change. Their common core lies in the encoding of causal force and resultant change, but they can be modeled as occupying different positions along a gradient of event construal, ranging from more concrete, process-oriented representations to more abstract, result-focused representations. This interpretation is consistent with Langacker’s (2008) view that linguistic structure constitutes a continuum of symbolic assemblies, in which more concrete and more abstract constructions are linked through gradual transitions rather than discrete boundaries. Although only two constructions are examined in the present study, this gradient is not limited to these endpoints but is assumed to include intermediate configurations that vary in the degree of displacement and result prominence. In a similar vein, Traugott & Trousdale (2013) argue that constructions exhibit gradience in synchronic grammar, with category membership being a matter of degree. Accordingly, the relation between the caused-motion and resultative constructions can be modeled as a continuum of semantically adjacent constructions linked by partial overlap, within which the two constructions occupy different positions. The observed priming asymmetry follows this gradient structure, with activation spreading more readily from the more concrete caused-motion construction to the more abstract resultative construction, reflecting differences in semantic accessibility. This

graded structure aligns with the view that linguistic representations are gradient in their degree of entrenchment (Schmid, 2020) and that semantic affinity shapes the strength of links across related constructions (Diessel, 2019).

Finally, regarding the potential metaphorical relationship between the caused-motion and resultative constructions, the unidirectional projection observed here aligns cognitively with Goldberg's (1995) metaphorical extension hypothesis, whereby activation flows from the concrete domain of spatial displacement to the abstract domain of state change. Crucially, the inhibitory effect observed at the complement position indicates a competitive relationship between the two constructions in how they encode results. This processing conflict can be partially interpreted as a reflection of the Unique Path Constraint (Goldberg, 1995), suggesting that a single patient within a single event cannot simultaneously accommodate two distinct types of resultative orientation. Thus, the present findings are consistent with a metaphorical association between the two constructions to a certain extent. At the same time, this interpretation does not necessarily entail that such a metaphorical relation must be modeled as a vertical inheritance link. As argued by Hong & Ma (2024), even constructions previously assumed to be related through metaphorical extension may exhibit horizontal links, and their bidirectional priming evidence suggests that metaphorical relatedness and horizontal connectivity are not mutually exclusive. Although the present study reveals unidirectional rather than bidirectional priming, this asymmetry can be interpreted as reflecting directional spreading within horizontal links shaped by prototypicality and semantic gradience, rather than requiring a derivational metaphorical inheritance relation. In a similar vein, recent constructional network research has emphasized that metaphorically related constructions may participate in multidimensional networks involving both taxonomic and horizontal associations, rather than being reducible to a single inheritance pathway (Sommerer & Smirnova, 2020). This view is also compatible with accounts of horizontal links as similarity-based associations whose strength may vary across related constructions, rather than as necessarily symmetric relations. Nevertheless, the present study does not attempt to determine the structural status of this metaphorical relation within the constructional

network. Instead, the observed asymmetry is interpreted at the processing level, reflecting differences in activation accessibility between related constructions within the same constructional family.

In summary, the unidirectional cross-constructional priming effect reflects the joint influence of prototypicality distribution and semantic similarity. As a high-prototypicality member, the caused-motion construction more easily spreads activation toward the semantically adjacent resultative construction. This finding supports viewing the *bǎ*-construction family as a dynamic multidimensional network, where connections are defined not only by abstract hierarchies but also by gradient distributions based on similarity.

5.3. The Role of Verb Repetition

This study investigated the role of verb repetition across various processing slots and different prime-target combinations within the basic frame NP₁ + *bǎ* + NP₂ + V + XP. The results indicate that the verb repetition effect demonstrates significant dependence on constructional conditions, specifically the pairing of the prime and target constructions.

At the verb position itself, the main effect of verb repetition did not reach statistical significance, indicating that lexical repetition did not consistently induce processing acceleration in the present study. This suggests that the influence of verb repetition at the verb presentation stage is limited and its impact on processing remains unstable. The observed finding is consistent with Mahowald et al.'s (2016) conclusion that lexical overlap is not a necessary condition for structural priming, but rather functions more as a variable, context-dependent enhancer. The absence of a significant effect at this position suggests a temporal dissociation between lexical priming and constructional integration. That is, even if the verb itself is easily retrieved, it does not necessarily facilitate the speaker's understanding of the constructional meaning. It further underscores the critical role of the interaction between the verb and the construction at the subsequent complement position.

At the complement first place, verb repetition exerted a significant modulatory effect on priming patterns. This indicates that lexical information does not exert uniformly influence throughout sentence processing. Instead, its impact emerges selectively at processing stages where constructional semantics must be explicitly realized and the event structure undergoes verification. Specifically, when the prime and target both involved the caused-motion construction, verb repetition significantly accelerated processing at the complement first word. This suggests successful verification of the event frame reinforced by the prime. Conversely, when the prime was caused-motion but the target was resultative, verb repetition amplified the pre-activated spatial-path expectation. This rendered the appearance of a resultative complement a prediction violation, thereby inducing processing delays. This co-occurrence of facilitation and inhibition does not represent a simple verb repetition effect but rather reflects how lexical information amplifies the consequences of successful or failed structural predictions within specific priming combinations. This pattern is consistent with prediction-violation accounts of processing cost in comprehension (Jaeger & Snider, 2013; Kuperberg, 2007; Levy, 2008; Ungerer, 2021), reflecting how lexical information amplifies the consequences of successful versus failed structural predictions. Notably, this pattern of co-occurring facilitation and inhibition emerged only under the caused-motion prime condition, with no similar effects observed under the resultative prime condition. This further indicates that the modulatory role of verb repetition is contingent upon the structural representations activated by the prime.

Overall, verb repetition does not constitute a necessary condition for cross-constructional priming, and its influence is highly dependent on processing slots and structural configurations. This finding aligns with the consensus in structural priming research that structural priming effects can stably emerge even when prime and target sentences do not share identical verbs or other lexical content (Bock & Griffin, 2000; Pickering & Branigan, 1998; Pickering et al., 2000). Lexical repetition, by contrast, functions primarily as a transient, conditional enhancer (Ferreira & Bock, 2006; Mahowald et al., 2016), and whether it affects structural priming depends on specific

processing stages and structural features.

Taken together, the present study suggests that verb repetition does not consistently produce facilitative effects at the lexical level across all processing stages, while its influence on structural processing is activated only under specific construction pairings and at specific critical slots. In the present study, no significant effect was found at the verb slot. Consistent with Ungerer (2021), however, the effects at structural integration regions were limited and largely condition-dependent rather than uniform. The present study further demonstrates that within the *bǎ*-construction network—where the complement serves as the critical differentiating element—the modulatory effect of verb repetition manifests more centrally as an amplifier of successful versus failed structural predictions. This discrepancy indicates that verb repetition does not simply generate a uniform positive boost. Rather, it interacts with semantic proximity between constructions and their centrality within the constructional network, thereby amplifying both facilitative and inhibitory effects under specific conditions.

6. Conclusion

This study introduces the maze task of self-paced reading paradigm into the investigation of Chinese syntactic processing, providing psycholinguistic evidence for the internal organization of the *bǎ*-construction family. The results reveal clear evidence for cross-constructional priming, but critically, this effect is asymmetric. Specifically, caused-motion primes significantly influenced the processing of resultative targets at the complement-first slot, yielding an inhibitory effect, whereas no corresponding effect was observed in the reverse direction. This asymmetry suggests that the relationship between the two constructions is not bidirectionally equivalent. Importantly, such a directional and inhibitory pattern is not expected under a simple superordinate–subordinate inheritance account, which would predict more generalized and facilitative activation across construction types. Instead, it supports a gradient organization within the constructional network, in which caused-motion appears to

function as a more strongly entrenched or prototypical representation that can influence the processing of semantically related constructions. The observed inhibition further indicates that priming may involve competition and prediction mismatch, rather than simple facilitation. In addition, verb repetition did not yield a stable facilitatory effect across processing stages, but instead showed context-dependent modulation at the complement region. This finding supports the view that structural priming operates independently of lexical overlap, while lexical repetition serves only a limited, stage-specific role. Taken together, these results suggest that cross-constructional priming in the *bǎ*-construction reflects structured relationships among constructions, shaped by both semantic similarity and differences in representational strength.

In addition, several limitations of the present study should be acknowledged. The maze-based self-paced reading task primarily captures online comprehension processes, and it remains to be determined whether the findings generalize to language production or more naturalistic reading contexts. Moreover, the present study focuses on only two sub-constructions within the *bǎ*-construction family, leaving other members and their interrelations unexplored. These limitations point to clear and productive directions for future research.

Future research may further combine the structural priming paradigm with other experimental methods to systematically examine the psychological reality of the metaphorical linkage proposed by Goldberg (1995) in Chinese. Although the direction of priming observed in the present study aligns cognitively with the metaphorical projection from spatial motion to change of state, this result does not provide direct evidence for an inheritance relation between these constructions. Instead, the observed unidirectional activation pattern more likely reflects horizontal links within the same constructional family—specifically, a unidirectional projection based on semantic mapping within similar event structure frames, rather than hierarchical inheritance in a strict sense. Subsequent studies could incorporate cross-domain priming, semantic judgment tasks, or neurocognitive measures to investigate whether constructional metaphors manifest as distinct types of processing effects or unfold over different temporal scales.

Moreover, future work may integrate large-scale corpus analyses to further validate the relationship between caused-motion and resultative constructions from a usage-based perspective. Examining their distributional patterns across semantic contexts, verb collocations, and discourse functions would help determine whether the semantic gradients observed experimentally are reflected in actual language use, thereby clarifying the relationship among semantic salience, frequency, and cognitive accessibility within constructional networks. The integration of experimental and corpus-based evidence will contribute to a more robust model of Chinese constructional organization.

Ethical statement: This study was approved by the ethics committee of the authors' institution (Approval date: 10 September 2024; Approval No. 202400002)

Appendix A. List of experimental items.

Type	Sentence	Category	Block
prime	小克把烟灰弹进烟灰缸	Caused-Motion	Block1
target	小赵把地毯卷至墙角	Caused-Motion	
prime	小天把奖杯举高了	Resultative	
target	小费把雕塑推回原处	Caused-Motion	
prime	小璟把纸片弹远了	Resultative	
target	小曹把牛奶倒入瓷缸	Caused-Motion	
prime	小帆把纸卷皱了	Resultative	
target	老石把画卷紧了	Resultative	
prime	小海把沙包抛远了	Resultative	
target	小西把帽子抛高了	Resultative	
prime	小达把椅子移近了	Resultative	
target	小杜把酒杯倒满了	Resultative	
prime	小石把镜头举高了	Resultative	
target	小彭把妈妈推醒了	Resultative	
prime	小秦把马桶砸破了	Resultative	
target	小朝把拳头砸向墙壁	Caused-Motion	Block2
prime	小藻把衣服挂进衣橱*	Caused-Motion	
target	小彩把档位挂入一档*	Caused-Motion	
prime	小丽把绳索抛进湖里	Caused-Motion	
target	小王把球鞋抛到观众席上	Caused-Motion	
prime	小特把球砸进球框	Caused-Motion	
target	小风把面霜擦到脸部	Caused-Motion	
prime	小张把小偷抓入监狱	Caused-Motion	
target	小然把绳子拉直了	Resultative	
prime	小波把信件丢进壁炉	Caused-Motion	
target	小溪把相机砸坏了	Resultative	
prime	小佩把风铃挂高了	Resultative	
target	小秀把蚊帐挂好了	Resultative	
prime	小松把可乐倒进茶杯	Caused-Motion	
target	小肯把杯子倒空了	Resultative	
prime	小应把钉子打进墙里*	Caused-Motion	Block3
target	小川把木桩打进土里*	Caused-Motion	
prime	小郝把灰尘弹干净了	Resultative	
target	小于把珠子弹往洞里	Caused-Motion	
prime	小河把冰糖打碎了	Resultative	
958	(continued)		
target	小娜把货物拉到仓库	Caused-Motion	

prime	老童把鼻涕擦到身上	Caused-Motion	
target	老丁把钢笔放入口袋	Caused-Motion	
prime	小麦把面包丢到江里	Caused-Motion	
target	小谢把问题丢远了	Resultative	
prime	小瑾把钞票卷进竹筒	Caused-Motion	
target	小栋把头发抓乱了	Resultative	
prime	小乔把屏幕擦坏了	Resultative	
target	小明把老人推到医院	Caused-Motion	
prime	小月把女儿扯进怀里	Caused-Motion	
target	小铭把男人扯到院内	Caused-Motion	
prime	小马把布拉平了	Resultative	
target	小李把顾客拉到景区	Caused-Motion	
prime	小福把衣服扯开了	Resultative	Block4
target	小佳把绳子扯断了	Resultative	
prime	小瑞把旗子举过头顶	Caused-Motion	
target	小红把窗户擦亮了	Resultative	
prime	小翁把袋子抓破了*	Resultative	
target	小鹏把母鸡抓进厨房*	Caused-Motion	
prime	老王把地漏移至角落	Caused-Motion	
target	小金把酒杯举到嘴边	Caused-Motion	
prime	小夫把花盆移到阳台	Caused-Motion	
target	小年把风扇移开了	Resultative	
prime	小许把架子放低了	Resultative	
target	小孙把剪刀丢坏了	Resultative	
prime	小丹把钥匙放到桌上	Caused-Motion	
target	小兰把花放远了	Resultative	
prime	小阎把饭碗打烂了	Resultative	
target	小兴把窗户推开了	Resultative	

(Note: Trials marked with an asterisk (*) were excluded from the final analysis to ensure lexical-semantic consistency and to prevent confounding from complement repetition. Specifically, these exclusions involve: (i) cases where the repeated verb forms denoted distinct meanings across constructions, and (ii) trials where the post-verbal complements were identical between prime and target, which could inflate the structural priming effect (Ungerer, 2021).)

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